Java Utilities

```
import java.util.Arrays;
package com.myorg; import software.amazon.awscdk.RemovalPolicy; import software.amazon.awscdk.Stack; import
software.amazon.awscdk.StackProps; import software.amazon.awscdk.services.dynamodb.*; import
software.constructs.Construct; import java.util.Collections;
```

Creating a Dynamo DB Table

Dependencies

```
<dependency>
    <groupId>software.amazon.awscdk</groupId>
    <artifactId>aws-cdk-lib</artifactId>
    <version>2.170.0</version>
</dependency>
```

```
TableV2 myTable = TableV2.Builder.create(this, "MyUserTable")
.tableName("users-table")
.partitionKey(Attribute.builder()
.name("userId")
.type(AttributeType.STRING)
.build())
.sortKey(Attribute.builder()
.name("createdAt")
.type(AttributeType.STRING)
.build())
```

To add a GSI

```
myTable.addGlobalSecondaryIndex(GlobalSecondaryIndexPropsV2.builder()
.indexName("email-index")
.partitionKey(Attribute.builder()
.name("email")
.type(AttributeType.STRING)
.build())
.projectionType(ProjectionType.ALL)
//.nonKeyAttributes(Arrays.asList("email", "status", "role"))
.build());
```

- Defining a sort key is optional
- What is projection type:
  - ALL
  - KEYS_ONLY
  - INCLUDE
- A PK+SK must be unique for every item. But GSI keys do not have to be unique, It will return all the items.
  gem

To query the table

Defining the Java class model

```
import software.amazon.awssdk.enhanced.dynamodb.mapper.annotations.*;

@DynamoDbBean
public class User {
private String userid;
private String createdAt;
private String email;
```

```
//
@DynamoDbPartitionKey
public String getUserid() { return userid; }
public void setUserid(String userid) { this.userid = userid; }
//
@DynamoDbSecondaryPartitionKey(indexNames = "createdAt-index")
public String getCreatedAt() { return createdAt; }
public void setCreatedAt(String createdAt) { this.createdAt = createdAt; }
//
public String getEmail() { return email; }
public void setEmail(String email) { this.email = email; }
}
```

Initialize the Client

```
DynamoDbClient ddbClient = DynamoDbClient.create();
DynamoDbEnhancedClient enhancedClient = DynamoDbEnhancedClient.builder()
.dynamoDbClient(ddbClient)
.build();
```

What is difference between a normal client and enhanced client?

Map physical table to generated name

```
DynamoDbTable<User> userTable = enhancedClient.table("users-table", TableSchema.fromBean(User.class));
```

Insert an Item

```
User newUser = new User();
newUser.setUserid("user_123");
newUser.setCreatedAt("2026-06-24T12:00:00Z");
newUser.setEmail("test@example.com");
userTable.putItem(newUser); // Inserts or overwrites entirely
```

Update an Item

```
User newUser = new User();
newUser.setUserid("user_123");
newUser.setCreatedAt("2026-06-24T12:00:00Z");
newUser.setEmail("test@example.com");
userTable.putItem(newUser);
```

Delete an Item

```
Key key = Key.builder().partitionValue("user_123").build();
userTable.deleteItem(key);
```

Query by Primary Key

```
Key key = Key.builder().partitionValue("user_123").build();
User user = userTable.getItem(key);
System.out.println("Found user: " + user.getEmail());
```

Query by GSI

```
import software.amazon.awssdk.enhanced.dynamodb.model.*;

// 1. Reference the GSI by its exact index name
DynamoDbIndex<User> createdAtIndex = userTable.index("createdAt-index");

// 2. Create the query conditional matching your GSI key value
QueryConditional queryConditional = QueryConditional
.keyEqualTo(Key.builder().partitionValue("2026-06-24T12:00:00Z").build());

// 3. Execute query and iterate over pages of results
SdkIterable<Page<User>> results = createdAtIndex.query(QueryEnhancedRequest.builder()
```

```java
.queryConditional(queryConditional)
.build());

results.forEach(page -> {
page.items().forEach(userItem -> {
System.out.println("User found via GSI: " + userItem.getUserid());
});
});
```